

Module 3: SQL Query Analysis

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This report contains all eleven questions, results, executable SQL, and explanations. The database snapshot contains 31,640 applicant entries. Queries use PostgreSQL through psycopg. The report was written in response to Part 4 of Module 3.

Question 1

How many entries are from applicants who applied for Fall 2026?

Result

Fall 2026 count	31,183
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Explanation

This query measures how many application entries belong to a chosen start term.

It first selects entries for Fall 2026, ignoring differences in capitalization.

It then counts the matching rows and returns a single total.

The total represents application entries, which may include repeat applicants.

SQL query

```
SELECT COUNT(*) AS fall_2026_count
FROM applicants
WHERE LOWER(term) = 'fall 2026';
```

Question 2

Among entries that provide a nationality classification, what percentage are international students?

Result

Percent international	46.13%
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Explanation

This query shows the share of entries classified as international students.

It first removes entries with a missing or empty nationality classification.

It divides the international count by all remaining entries and multiplies by 100.

The result is rounded to two decimal places; an empty group avoids division by zero.

SQL query

```
SELECT ROUND(
  100.0 * COUNT(*) FILTER (
    WHERE LOWER(us_or_international) = 'international'
  ) / NULLIF(COUNT(*), 0), 2
) || '%' AS percent_international
FROM applicants
WHERE us_or_international IS NOT NULL
AND us_or_international <> '';
```

Question 3

What are the average GPA, GRE Quantitative, GRE Verbal, and GRE Analytical Writing scores of applicants who provide each metric?

Result

Average gpa	3.79
Average gre quantitative	260.11
Average gre verbal	161.52
Average gre analytical writing	8.75

Explanation

This query summarizes reported academic performance using four separate averages. Each average uses the available values for that measure and ignores missing values. An entry can contribute to one average even if its other scores are missing. Results are rounded to two decimal places; unusual scores are included as reported.

SQL query

```
SELECT
  ROUND(AVG(gpa)::numeric, 2) AS average_gpa,
  ROUND(AVG(gre)::numeric, 2) AS average_gre_quantitative,
  ROUND(AVG(gre_v)::numeric, 2) AS average_gre_verbal,
  ROUND(AVG(gre_aw)::numeric, 2) AS average_gre_analytical_writing
FROM applicants;
```

Question 4

What is the average GPA of American applicants who applied for Fall 2026?

Result

Average gpa	3.79
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Explanation

This query summarizes GPA for a group defined by nationality and start term. It selects American applicants for Fall 2026 who have a reported GPA. It adds their GPA values and divides by the number of entries with a GPA. The result is one group average, rounded to two decimal places.

SQL query

```
SELECT ROUND(AVG(gpa)::numeric, 2) AS average_gpa
FROM applicants
WHERE LOWER(term) = 'fall 2026'
      AND LOWER(us_or_international) = 'american'
      AND gpa IS NOT NULL;
```

Question 5

What percentage of Fall 2025 entries with a reported status are acceptances?

Result

Acceptance percentage	47.39%
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Explanation

This query measures how often entries with a reported outcome show acceptance.

It selects Fall 2025 entries and removes missing or empty statuses.

It divides accepted entries by that total and converts the ratio to a percentage.

The result is rounded to two decimal places, with protection against division by zero.

No Fall 2025 statuses are missing here, so the total includes every entry for that term.

SQL query

```
SELECT ROUND(  
    100.0 * COUNT(*) FILTER (WHERE LOWER(status) = 'accepted')  
    / NULLIF(COUNT(*), 0), 2  
) || '%' AS acceptance_percentage  
FROM applicants  
WHERE LOWER(term) = 'fall 2025'  
    AND status IS NOT NULL  
    AND status <> '';
```

Question 6

What is the average GPA of accepted applicants who applied for Fall 2026?

Result

Average gpa	3.78
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Explanation

This query summarizes the reported GPA of applicants who received admission offers. It selects accepted Fall 2026 entries and excludes those without a GPA. It calculates the mean GPA of the remaining entries and rounds to two decimal places. The result describes this admitted group, rather than a minimum admission requirement.

SQL query

```
SELECT ROUND(AVG(gpa)::numeric, 2) AS average_gpa
FROM applicants
WHERE LOWER(term) = 'fall 2026'
      AND LOWER(status) = 'accepted'
      AND gpa IS NOT NULL;
```

Question 7

How many entries are from applicants who applied to Johns Hopkins University for a master's degree in Computer Science?

Result

Jhu cs masters count	8
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Explanation

This query counts entries matching a university, subject, and degree level.

It searches the combined program field for Johns Hopkins or JHU and Computer Science.

It also requires a degree label beginning with “master, ignoring capitalization.

All matching entries are counted, regardless of their start term or admission outcome.

SQL query

```
SELECT COUNT(*) AS jhu_cs_masters_count
FROM applicants
WHERE (
    program ILIKE '%Johns Hopkins%'
    OR program ~* '\mJHU\M'
)
AND program ILIKE '%Computer Science%'
AND degree ILIKE 'master%';
```

Question 8

How many Fall 2026 entries are acceptances for a PhD in Computer Science at Georgetown, MIT, Stanford, or Carnegie Mellon, using the original downloaded fields?

Result

Original field count	30
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Explanation

This query counts entries that meet several admission and program conditions together. It selects accepted Fall 2026 Computer Science PhD entries at the four named universities. It identifies the subject and university within the original combined program field. Flexible text matching handles name variations; whole-word MIT avoids matching RMIT.

SQL query

```
SELECT COUNT(*) AS original_field_count
FROM applicants
WHERE term ILIKE 'Fall 2026'
      AND status ILIKE 'Accepted'
      AND degree ILIKE 'PhD'
      AND program ILIKE '%Computer Science%'
      AND (
        program ILIKE '%Georgetown%'
        OR program ILIKE '%Massachusetts Institute of Technology%'
        OR program ~* '\mMIT\M'
        OR program ILIKE '%Stanford%'
        OR program ILIKE '%Carnegie Mellon%'
      );
```

Question 9

Repeat Question 8 using the LLM-generated university and program fields. Report the original count, LLM count, and their difference.

Result

Original field count	30
Llm field count	30
Difference	+0

Explanation

This query compares counts obtained from original and LLM-generated program details. It first selects a shared group of accepted Fall 2026 PhD entries. It counts subject and university matches in each version, then subtracts original from LLM. Both counts are 30, and they are the same 30 entries, because the query matches substrings — the university name is found just as reliably inside the original combined program field as it is in the separate `llm_generated_university` column. The standardized fields only prove their worth under exact matching (see pg 3 of screenshots.pdf);

Question 9 (continued)

SQL query

```
WITH eligible AS (  
    SELECT program, llm_generated_program, llm_generated_university  
    FROM applicants  
    WHERE term ILIKE 'Fall 2026'  
        AND status ILIKE 'Accepted'  
        AND degree ILIKE 'PhD'  
) , counts AS (  
    SELECT  
        COUNT(*) FILTER (  
            WHERE program ILIKE '%Computer Science%'  
            AND (  
                program ILIKE '%Georgetown%'  
                OR program ILIKE '%Massachusetts Institute of Technology%'  
                OR program ~* '\mMIT\M'  
                OR program ILIKE '%Stanford%'  
                OR program ILIKE '%Carnegie Mellon%'  
            )  
        ) AS original_field_count,  
        COUNT(*) FILTER (  
            WHERE llm_generated_program ILIKE '%Computer Science%'  
            AND (  
                llm_generated_university ILIKE '%Georgetown%'  
                OR llm_generated_university ILIKE '%Massachusetts Institute of Technology%'  
                OR llm_generated_university ~* '\mMIT\M'  
                OR llm_generated_university ILIKE '%Stanford%'  
                OR llm_generated_university ILIKE '%Carnegie Mellon%'  
            )  
        ) AS llm_field_count  
    FROM eligible  
)  
SELECT original_field_count, llm_field_count,  
        llm_field_count - original_field_count AS difference  
FROM counts;
```

Question 10

What was the lowest reported GPA of an applicant admitted to a doctoral program at each of the four universities from Question 8 for Fall 2026?

Result

University	Minimum GPA
Carnegie Mellon University	3.13
Georgetown University	3.52
Massachusetts Institute of Technology	3.40
Stanford University	3.00

Explanation

This query finds the lowest reported GPA among admitted doctoral entries at each university. It selects Fall 2026 PhD acceptances with a GPA, including every subject area. It maps matching university names to four standard labels and groups entries by university. It returns the smallest GPA in each group; these values are not admission requirements.

SQL query

```
WITH admitted AS (  
  SELECT  
    CASE  
      WHEN llm_generated_university ILIKE '%Georgetown%'  
        THEN 'Georgetown University'  
      WHEN llm_generated_university ILIKE '%Massachusetts Institute of Technology%'  
        OR llm_generated_university ILIKE 'MIT'  
        THEN 'Massachusetts Institute of Technology'  
      WHEN llm_generated_university ILIKE '%Stanford%'  
        THEN 'Stanford University'  
      WHEN llm_generated_university ILIKE '%Carnegie Mellon%'  
        THEN 'Carnegie Mellon University'  
    END AS university,  
    gpa  
  FROM applicants  
  WHERE status ILIKE 'Accepted'  
    AND term ILIKE 'Fall 2026'  
    AND degree ILIKE 'PhD'  
    AND gpa IS NOT NULL  
)  
SELECT university, MIN(gpa) AS lowest_accepted_gpa  
FROM admitted  
WHERE university IS NOT NULL  
GROUP BY university  
ORDER BY university;
```

Question 11

What were the lowest reported GPAs of international versus American applicants admitted to doctoral programs at each of the four universities from Question 8 for Fall 2026?

Result

University	American	International
Carnegie Mellon University	3.56	3.13
Georgetown University	3.52	3.73
Massachusetts Institute of Technology	3.40	3.70
Stanford University	3.00	3.60

Explanation

This query compares the lowest reported GPAs of two nationality groups at each university. It selects Fall 2026 PhD acceptances with a GPA across all subjects at the four universities. Within each university, it finds separate minimum GPAs for American and international entries. Other classifications are excluded; a missing result means no available GPA, not a GPA of zero.

SQL query

```
WITH admitted AS (
  SELECT
    CASE
      WHEN llm_generated_university ILIKE '%Georgetown%'
        THEN 'Georgetown University'
      WHEN llm_generated_university ILIKE '%Massachusetts Institute of Technology%'
        OR llm_generated_university ILIKE 'MIT'
        THEN 'Massachusetts Institute of Technology'
      WHEN llm_generated_university ILIKE '%Stanford%'
        THEN 'Stanford University'
      WHEN llm_generated_university ILIKE '%Carnegie Mellon%'
        THEN 'Carnegie Mellon University'
    END AS university,
    us_or_international, gpa
  FROM applicants
  WHERE status ILIKE 'Accepted'
    AND term ILIKE 'Fall 2026'
    AND degree ILIKE 'PhD'
    AND gpa IS NOT NULL
    AND (us_or_international ILIKE 'American'
      OR us_or_international ILIKE 'International')
)
SELECT university,
  MIN(gpa) FILTER (WHERE us_or_international ILIKE 'American')
  AS lowest_american_gpa,
  MIN(gpa) FILTER (WHERE us_or_international ILIKE 'International')
  AS lowest_international_gpa
FROM admitted
WHERE university IS NOT NULL
GROUP BY university
ORDER BY university;
```